

(Mis)Understanding Numbers

THERE IS AN old joke about a senior partner interviewing Certified Public Accountants for a junior position. The interview consists of a single question: “How much is $2 + 2$?” The CPA who gets hired? The one who responds, “How much do you want it to be?”

Numbers can be precise, and numbers can mislead. Like any tool, the result they produce is in the hands of the craftsman wielding them. Their apparent precision can easily fool us.

To avoid math trickery and abuse that befalls too many investors, you need to understand a few simple concepts. We can do this with some familiarity of basic mathematics. Just high school level basics, including a tiny bit about probabilities and even less about statistics.

You do not need a lot of math to be a good investor. The kinds of things that most investors need to do are look at the relative performance of a fund, or the after-tax equivalent yield on municipal bonds, earnings, or dividends yield for individual companies. There are online calculators that make this fast and easy to do.

The bigger and more challenging task is to avoid being fooled by malicious actors who are facile with math. They abuse numbers to scare you out of markets, to make you fearful about the future, to mislead you about stocks, bonds, economics, politics, just about anything. They can create doubt in your mind about your plan.

The same is true with economic data: Models used to determine nonfarm payrolls or changes in the Consumer Price Index mean that these data series are very noisy, with a wide margin of error. A few insights will prevent you from being fooled by the endless fire hose of economic news. On a long-term trend line, most of it is meaningless noise.

I’ll share multiple examples of how we can be deceived with numbers: